



D. Fusi

Linked Open Data in Digital Scholarly Editions

SunoikisisDC Digital Scholarly
Editions in Classics, Byzantine,
and Medieval Studies

Summer 2026





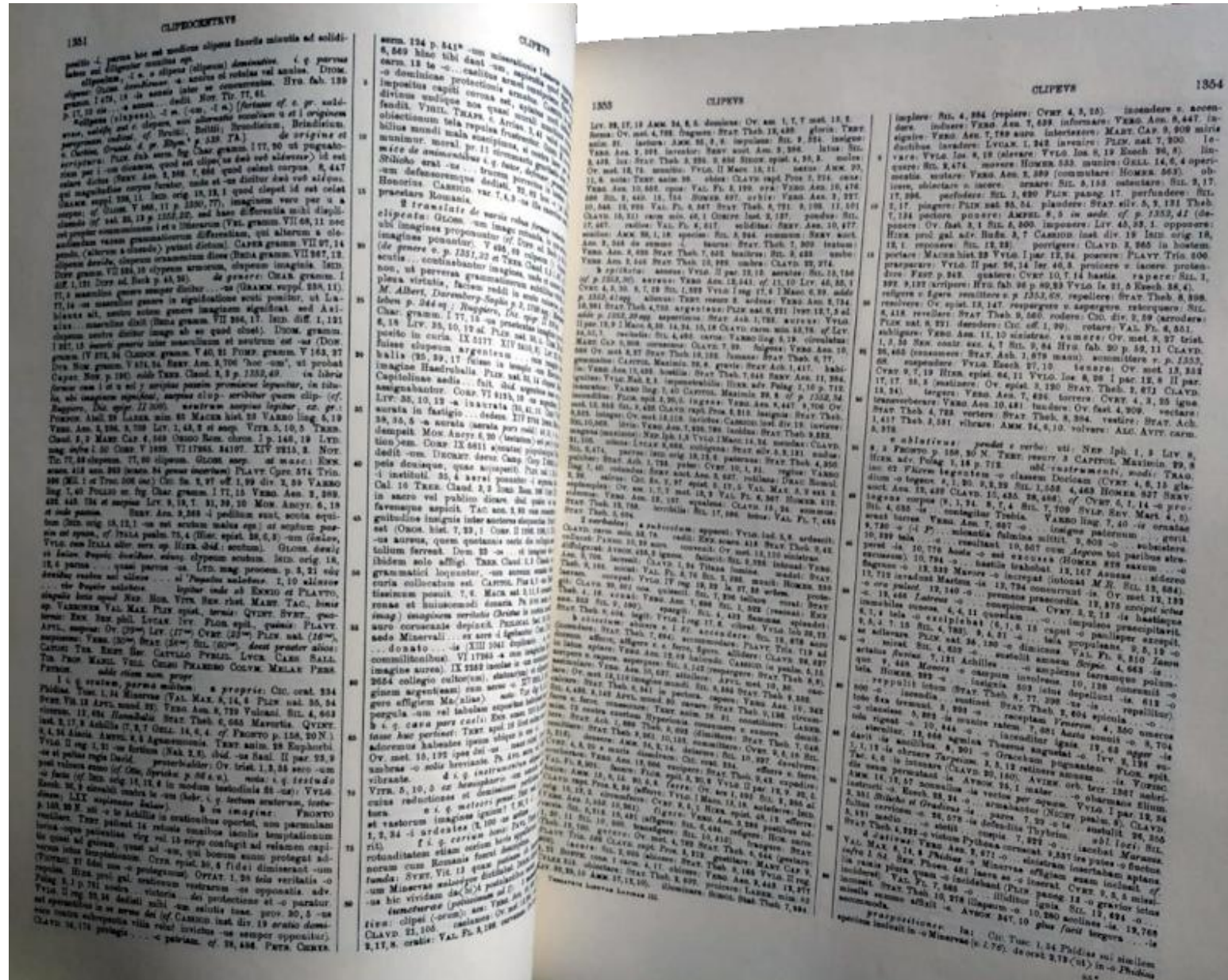
Modeling

Shaping Digital Content



Structured Data: an Example

- pre-digital word
- printed dictionary:
only for humans



PDF Dictionary

- digitized from paper
- search limited to OCR text:
 - paper-based layout
 - extremely noisy
 - faulty or missing

725

ADLVC.

ADMINIC

Adluc. *nomen barbaricum.* CORP. XIII 1177 (*in agro Pictonum, litteris saec. I*) C. Rufinio Iul(ius) Adluc. her(es). Adluc(us?) *Hirschfeld, Adluc(tus?) Holder, Altcelt. Sprachsch. I p. 42.* Otto.

Adlucca, -ae *nom. mul. peregr.* CORP. V 4536 (*Brixiae*) Arugus Iovincilli f. sib. et Adlucae (-lu- non vidit *Mommsen*) Messili f. uxo(ri).

? **Adma**<... *nomen peregrinum.* CORP. III 11513 (*lapidem detritum descr. Hirschfeld, Viruni*) d. m. Counerti Ituci / Adma<///> in<t?>ii *Gemellinus si<bi ///> f. c.* Diehl.

Admaeus *nom. gent.* CORP. VI 6908 Admaeo Peculiari Memmius Eutactus fec. fil. suo.

Admagetobriga, -ae. *locus in Gallia, ubi Ariovistus Haeduos vicit.* cf. *Ihm, PW. I 377.* CAES. Gall. 1, 31, 12 quod proelium factum sit Admagetobrigae (admagetobrige, admagetobriae, (admagetobrio) *codices optimi*; ad magetobriam *Vindobon. I.* Otto.

admallo, -ātus, -āre. *ab ad et mallum (i. dies dictus). terminus iuris germanici. in ius vocare:* LEX Sal. 50, 2 creditor debet eum eum admallare... et sic mallare debet (52). 50, 3 quem legitime habeo... admallatum (51, 1 bis). 53, 1 si quis ad hineum ('Kesselfang') admallatus fuerit (53, 5). *ibid.* qui admallatus est (56). 56 ille qui admallatus est. *ibid.* ille qui admallat. *ibid.* si... qui admallatus est ad nullum placitum venire voluerit. capit. 1, 9 vos admallo, ut in mallo proximo veniatis. extrav. A 2 si eum admalluerit (*sic in cod. H.*) cf. *Kalb, Roms Juristen p. 128.*

admandat NOT. Tir. 30, 64 post mandat amandat.

admanet NOT. Tir. 30, 40 post manet amanet. 30, 46 admansit post mansit amansit.

(admanuēnsis) *vox ficta ab ad manum secundum amanuens.* CASSIAN. conl. 19, 4, 2 cibus... in prochirio id est admanuensi sporta reponitur. cf. GLOSS. III 438, 19 ad manum ἐν προχίρῳ.

admanumdeductor, -ōris *m.* ITALIA cod. Cantabr. act. 13, 11 quaerebat admanumdeductores (χειραγωγούς; VVLG. qui ei manum daret).

[**admartyrizo,** -āre] *verbum in lexica inlatum ex COMM. instr.* 2, 17, 19 si refrigerare cupis, ad martyres i (*sic recte Ludwig, ad martyrisi codd., admartyriza edd. vulgo.* V.

mundum quae admensa sororis. p. 72, 12 Caesar... censuit omnem simos viros (cf. IVL. HONOR. geogr. MART. CAP. 6, 588 permeatam crel amensum cod. Bern. 8, 829.

Admētus, -ī, Ἀδμήτος. CORP. Addmeto. DRAC. Romul. 8, 210 A 1, 22 *ridicule interpretatus est:* Ad adire non poterit metus. cf. MYTH. Admeto regi Thessaliae (VAL. M. Pheretis filius (173 p. 28, 16) ma Thessalia monte Chalcodonio... hu LYGD. 3, 4, 67 me quondam Admeti (TIB. 2, 3, 11). OV. ars 2, 239 Cynthi fertur (cf. SEN. Herc. f. 451). VAL. fuderat arcu. STAT. Theb. 5, 434 Admetum (cf. SCHOL. ad l.). TE cf. apol. 14 (MIN. FEL. 23, 5 CYP civ. 18, 13 cum Hercule). HYG. f. 20 PROB. Verg. georg. 3, 1 MACR. sat VIB. SEQ. geogr. p. 145, 18 Amphry pecus fertur pavisse (SERV. Aen. 7, 2 CVLEX 262 Alcestis ab omni invio riti... Admeti fata (*sic Bemb.*) m 25 meti coniunx (OV. Pont. 3, 1, 106). 4, 6, 1. STAT. Theb. 6, 380 datur mori (cf. SCHOL. ad l.). silv. 5, 3, (sc. reversus ex Averno). cf. 3, 3, 1 redimens Pheraei uxor. HYG. fab. 30 Iov. 1, 45. ad rem v. HYG. fab. MYTHOGR. 1, 92, 2, 154. OV. Ib lias). CORP. V 8265 in arca Ag batus pedit innixus et mulier vela Alc(estis). STAT. Theb. 6, 332 35 vix steriles compescit equas. 6, 37 Amphiarum Thessalus aequat eu

HTML Dictionary

- **presentational** markup designed for hypertext layout: paragraphs, titles, emphasis, table...
- a **lemma** might be bold when this is the rendition for ; but “bold” does not mean “lemma”, and styles might change
- full text **search**

ABACTOR Ab*ac"tor, n. Etym: [L., fr. abigere to drive away; ab+agere to drive.] (Law)

Defn: One who steals and drives away cattle or beasts by herds or droves. [Obs.]

ABACULUS

A*bac"u*lus, n.; pl. Abaculi. Etym: [L., dim. of abacus.] (Arch.)

Defn: A small tile of glass, marble, or other substance, of various colors, used in making ornamental patterns in mosaic pavements. Fairholt.

ABACUS

Ab"a*cus, n. E. pl. Abacuses ; L. pl. Abaci. Etym: [L. abacus, abax, Gr.

1. A table or tray strewn with sand, anciently used for drawing, calculating, etc. [Obs.]
2. A calculating table or frame; an instrument for performing arithmetical calculations by balls sliding on wires, or counters in grooves, the lowest line representing units, the second line, tens, etc. It is still employed in China.
3. (Arch.) (a) The uppermost member or division of the capital of a column, immediately under the architrave. See Column. (b) A tablet, panel, or compartment in ornamented or mosaic work.
4. A board, tray, or table, divided into perforated compartments, for holding cups, bottles, or the like; a kind of cupboard, buffet, or sideboard. Abacus harmonicus (Mus.), an ancient diagram showing the structure and disposition of the keys of an instrument. Crabb.

ABADA

Ab"
D

<p>ABACUS
 Ab"a*cus, n. E. pl. Abacuses ; L. pl. Abaci. Etym: [L. abacus, abax,
 Gr.</p>

<p>1. A table or tray strewn with sand, anciently used for drawing, calculating, etc. [Obs.]</p>

<p>2. A calculating table or frame; an instrument for performing arithmetical calculations by balls sliding on wires, or counters in grooves, the lowest line representing units, the second line, tens, etc. It is still employed in China.</p>

XML Dictionary

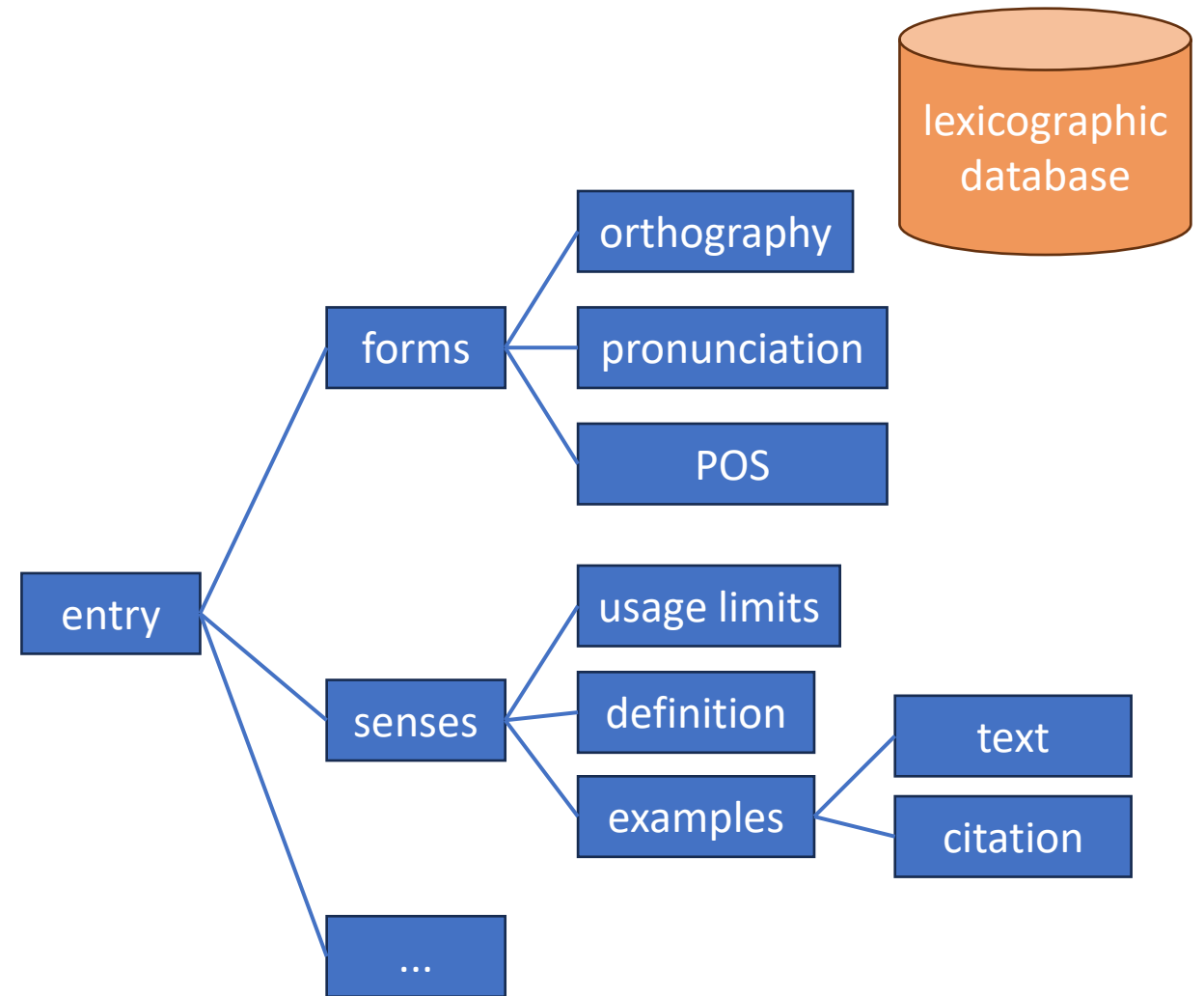
TEI Dictionaries Guidelines Example

- **semantic** markup designed for dictionary text
- a **lemma** is marked as a specific orthography (**orth**) inside a form, adding its hyphenation (**hyph**), pronunciation (**pron**), POS (**pos**), and definition (**def**)
- portions of text can be indexed for more powerful combined **searches**
- **presentations** just come from transforming XML into HTML

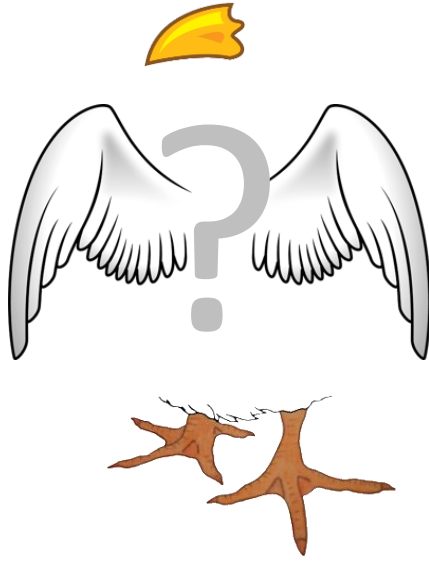
```
<entry>
  <form>                                orthography
    <orth>competitor</orth>             hyphenation
    <hyph>com|peti|tor</hyph>
    <pron>k@m"petit@(r)</pron>          pronunciation
  </form>
  <gramGrp>
    <pos>n</pos>                        part of speech (noun)
  </gramGrp>                           definition
  <def>person who competes.</def>
</entry>
```

DB Dictionary

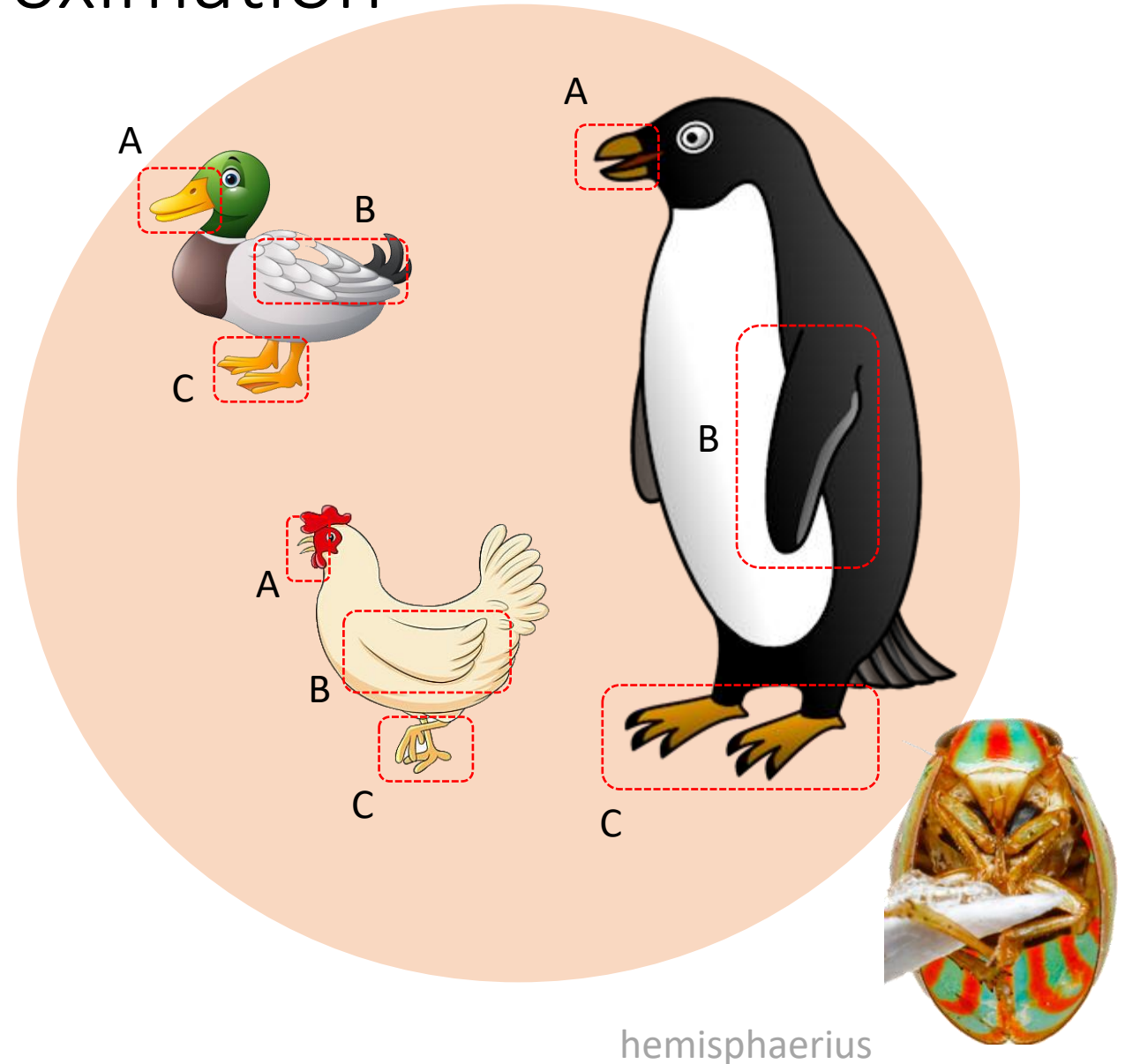
- on this way, one might also drop the text about lexical data and just fill an ideal lexical “**form**” in a database
- we might have **entries**, which include forms, each with graphical value, phonological value, POS...
- no longer a text **talking** about a word, but a set of data
- fast **search, filtering, analytics, linking...**



Data Modeling and Approximation

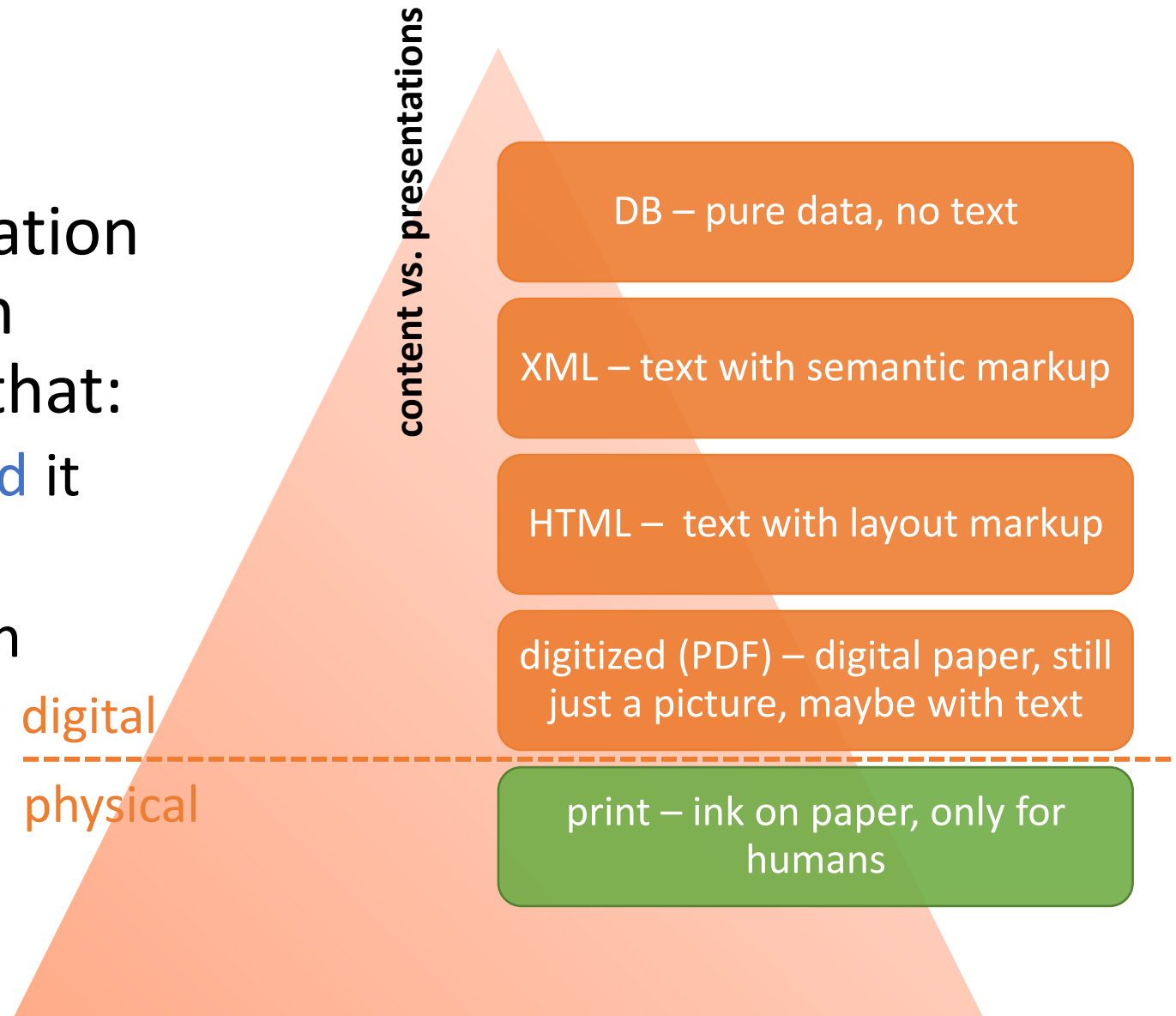


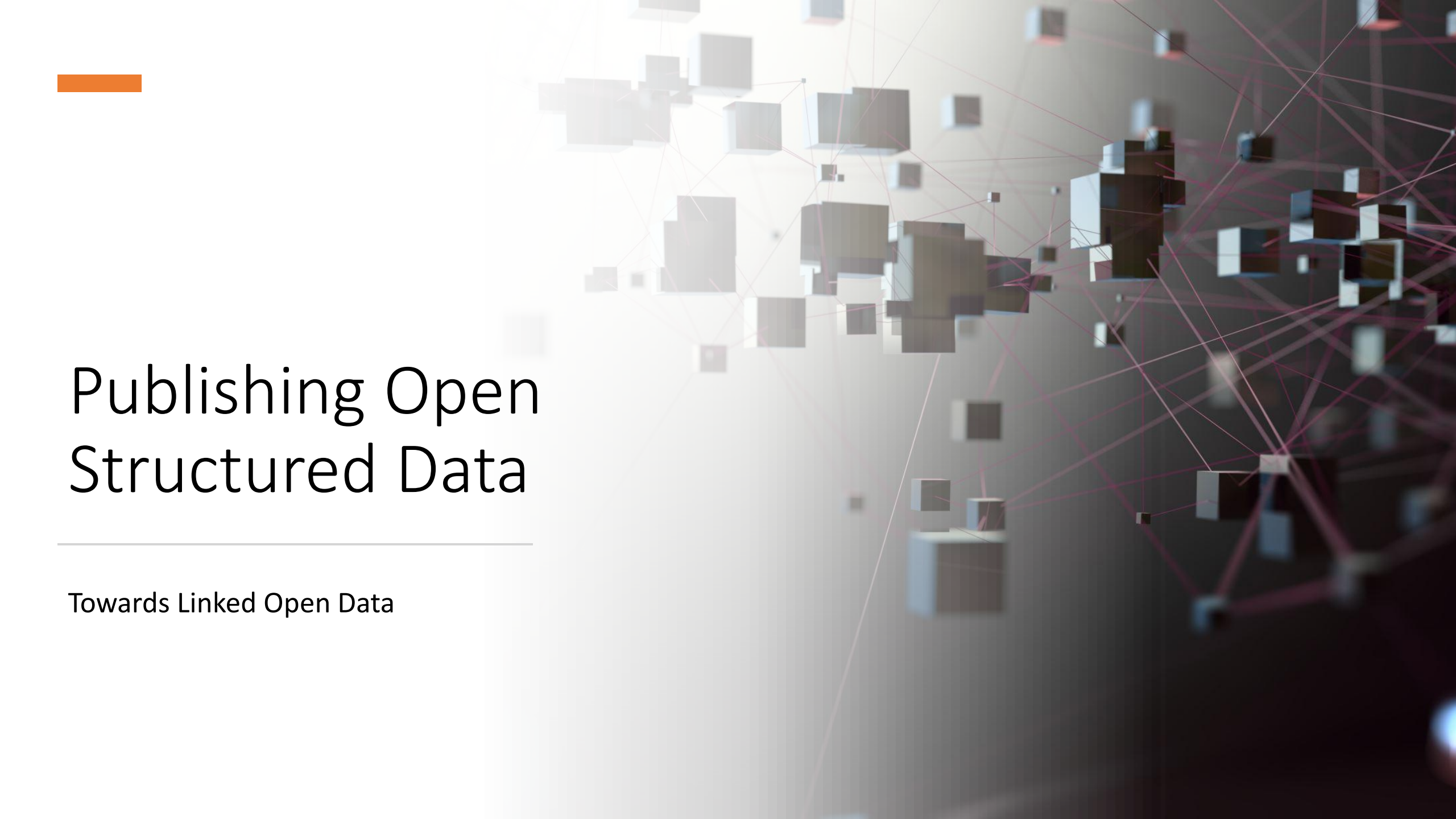
selecting features relevant
to our vision and intent



Structured Data

- structured data is information **organized** according to an explicit formal **model** so that:
 - computers can **understand** it
 - systems can **process** it
 - interfaces can **present** it in different ways



The background of the slide features a complex, abstract network of interconnected cubes and lines. The cubes are rendered in various shades of gray, white, and dark blue, creating a sense of depth and dimension. They are connected by a web of thin, reddish-pink lines that crisscross the frame, suggesting a global or interconnected data network. The overall aesthetic is modern and technological.

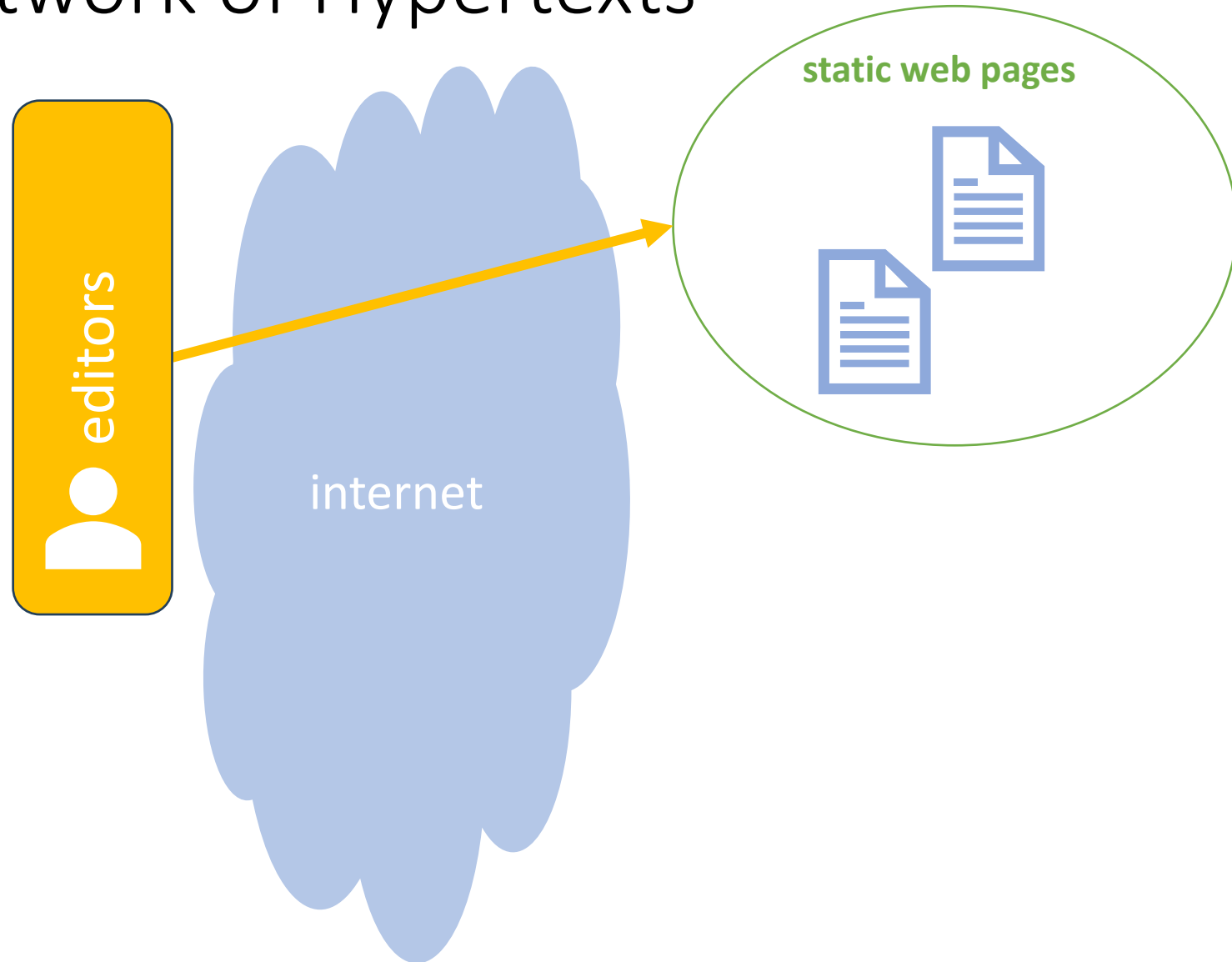
Publishing Open Structured Data

Towards Linked Open Data

Web 1.0: a Network of Hypertexts

WEB 1.0

web of hypertexts

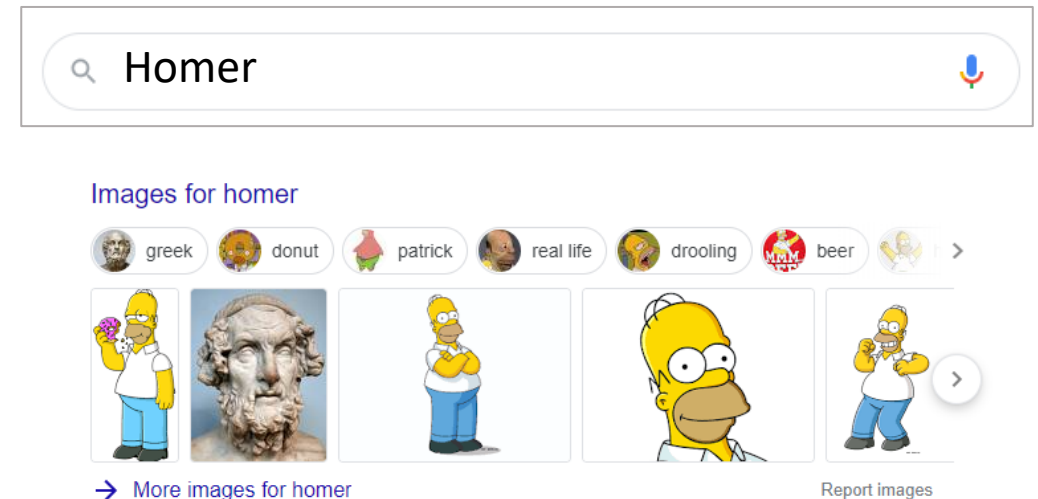


Web Roots and Data Mining

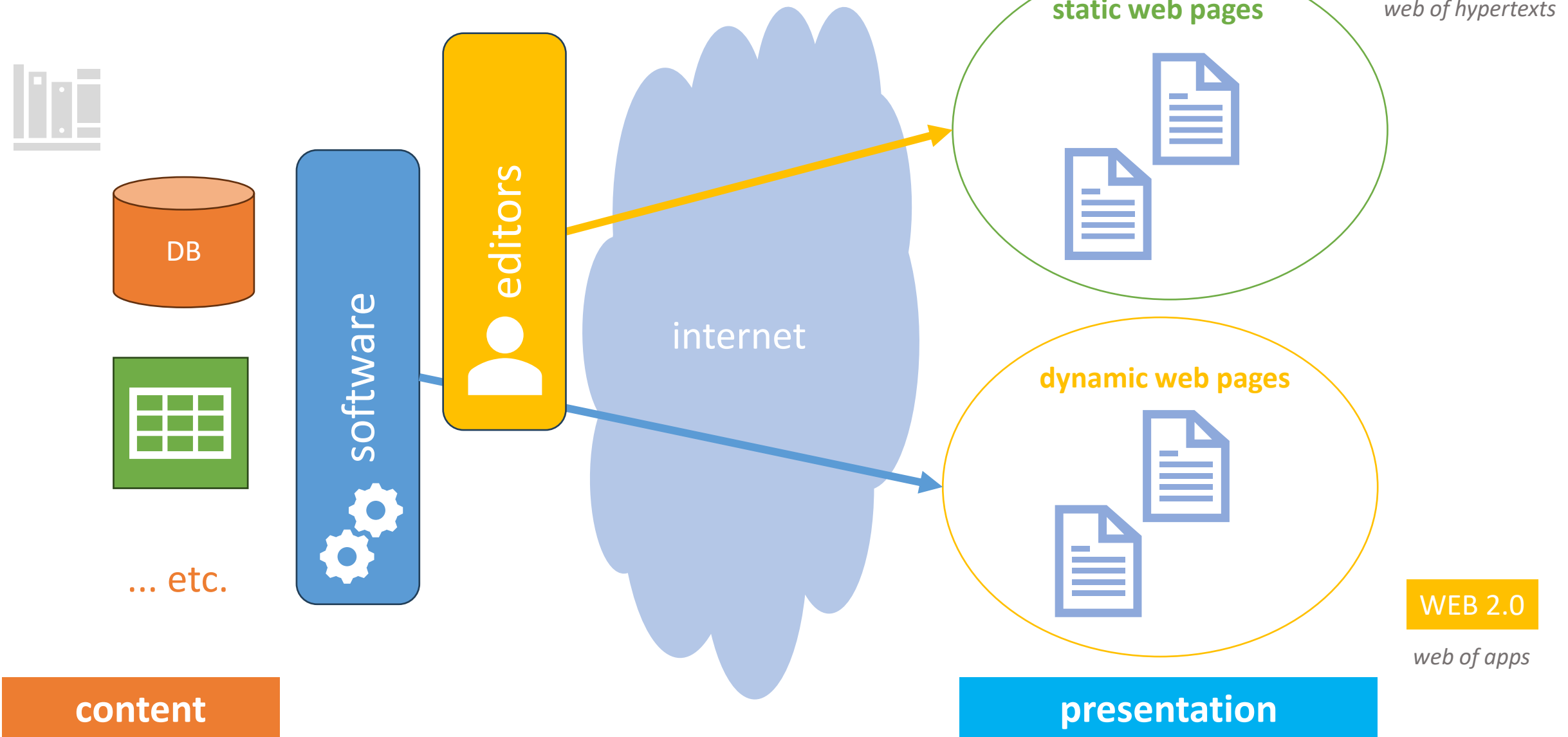
- **search engine**: like an **index** to a huge book (the web of hypertexts)
- each **hypertext** essentially has:
 - **text** in a natural language, intended for human readers
 - **links** to other hypertexts
- a search engine indexes normalized text, and takes into account a number of related factors, like incoming links

the human language is **ambiguous**

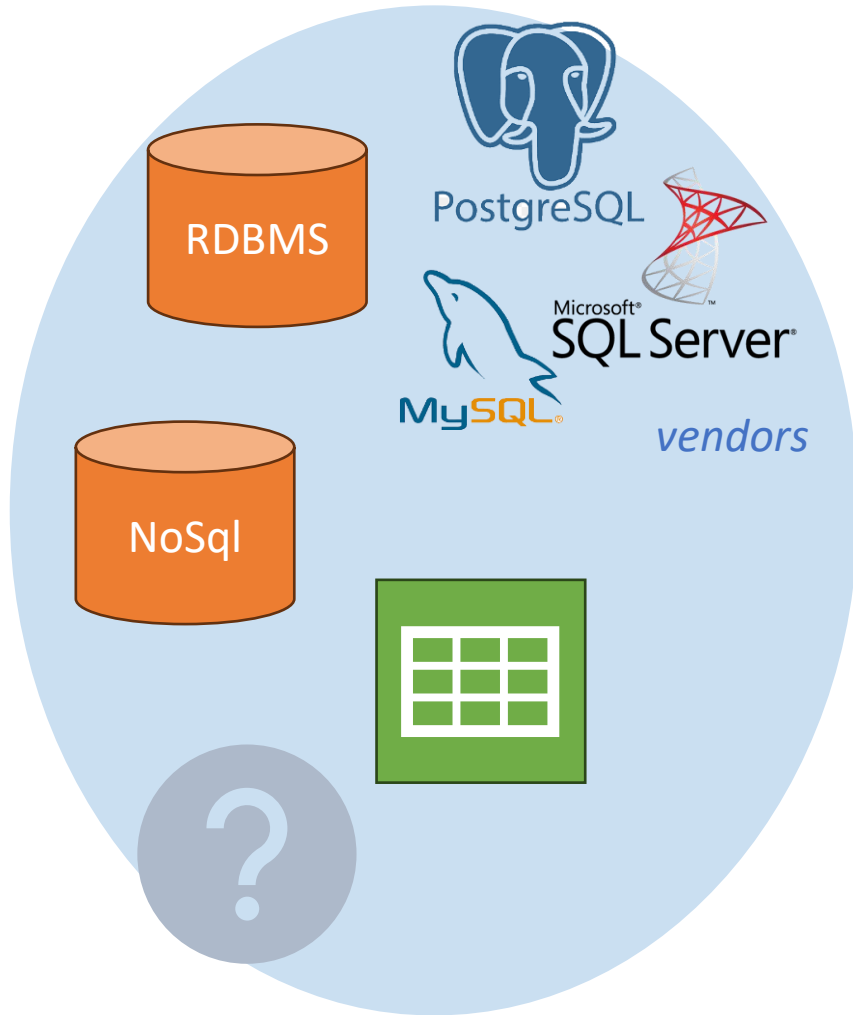
human languages are **different**



Wen 2.0 – a Web of Apps



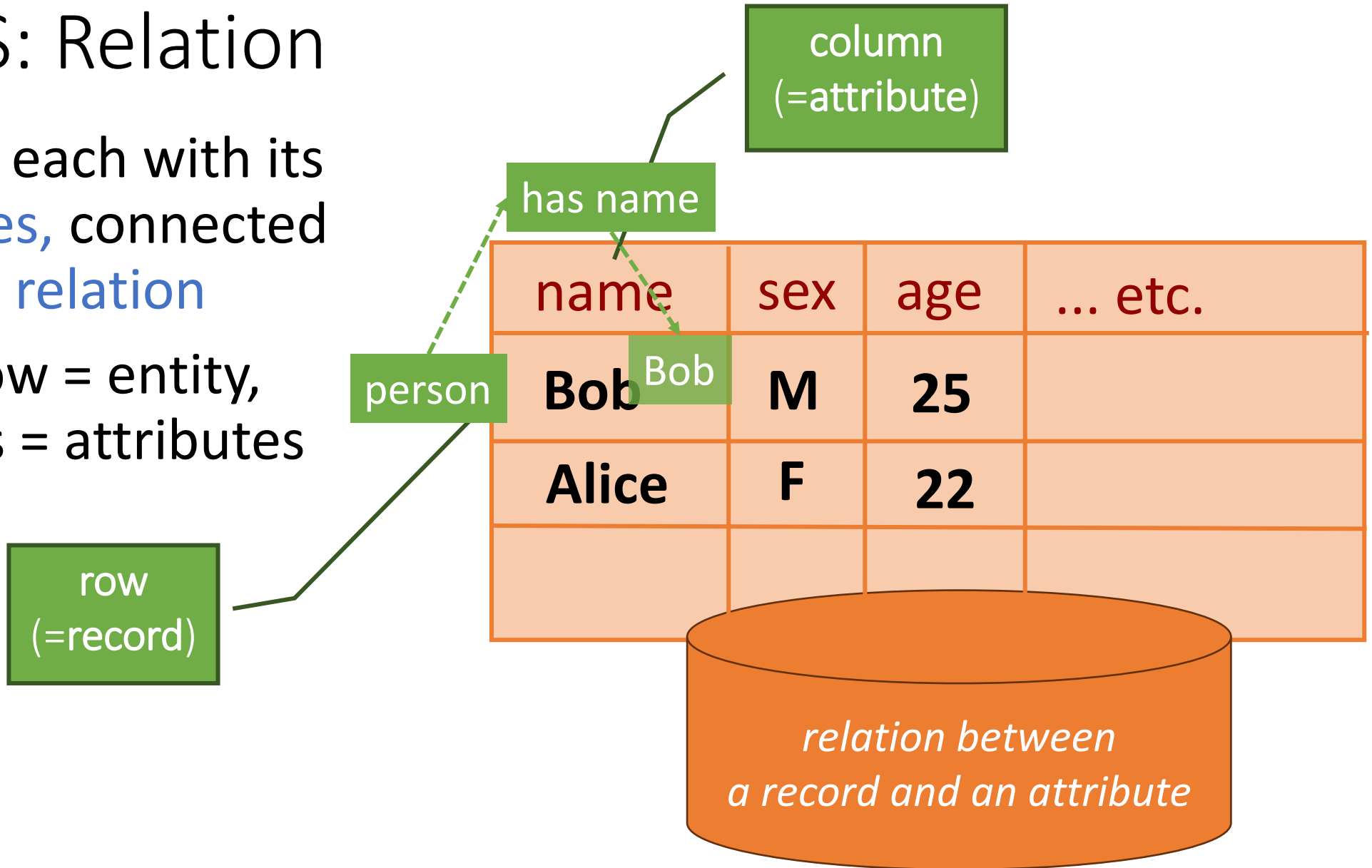
A World of Data Cages



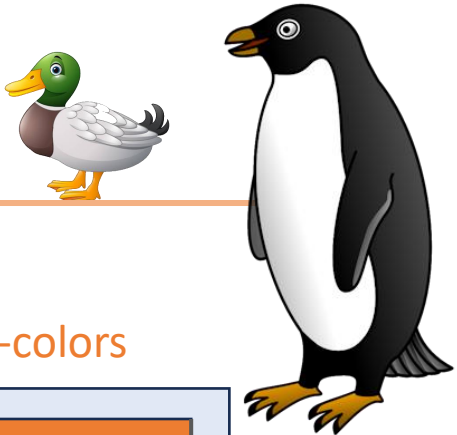
- various database **types** and **vendors** based on same abstraction
- incompatible **binary formats** though most share abstractions
- own data **models**
- software **tailored** to each combination of these factors

RDBMS: Relation

- **entities**, each with its **attributes**, connected to it in a **relation**
- **table**: row = entity, columns = attributes



Different Schemas



attributes of each bird

id	size	shape	genus	species
		bird 1		
		bird 2		

connection birds-colors

bird_id	color_id
	connection 1
	connection 2

attributes of each bird's name

id	bird_id	name	language
		name 1	
		name 2	

attributes of each color

id	color
	color 1
	color 2

Data Distribution: Tabular Data Sample



list of persons:

column	column	column	column
ID	name	birthDate	birthPlace
156	Marco Polo	1254	Venice
1201	Niccolò Polo	1230	Venice
368	Maffeo Polo	1230	Venice

row

row

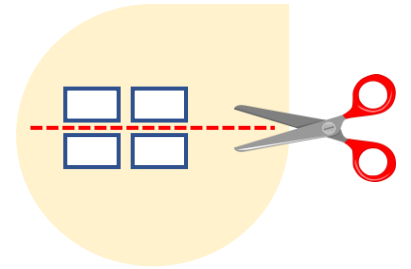
row

*data records, each
row is a person*

metadata

*meaning of each cell in
a row of data*

Distributing by Rows



ID	name	birthDate	birthPlace
56	Marco Polo	1254	Venice

server A



ID	name	birthDate	birthPlace
12	Niccolò Polo	1230	Venice
35	Maffeo Polo	1230	Venice

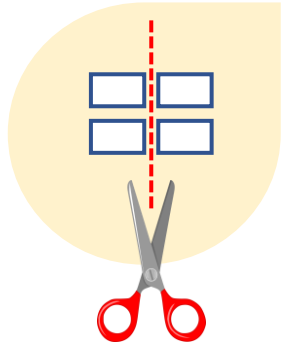
server B



duplicate
metadata

*to define the meaning
of each column in rows*

Distributing by Columns



ID	name
56	Marco Polo
12	Niccolò Polo
35	Maffeo Polo



server A

ID	birthDate	birthPlace
56	1254	Venice
12	1230	Venice
35	1230	Venice

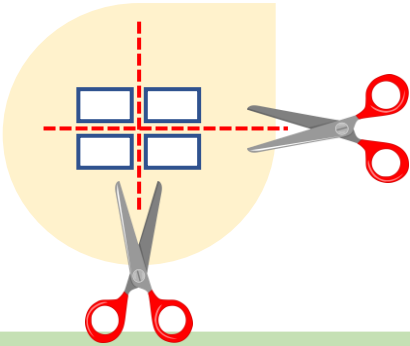


server B

duplicate
IDs

to link columns to the same row (person)

Distributing by **both** Columns and Rows (**Cells**)



ID	name
56	Marco Polo

ID	name
12	Niccolò Polo

ID	name
35	Maffeo Polo

server A

ID	birthDate	ID	birthPlace
56	1254	56	Venice

ID	birthDate	ID	birthPlace
12	1230	12	Venice

server B

1 identifier

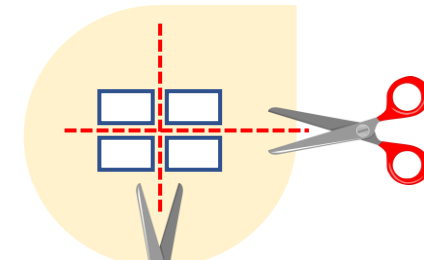
2 birthPlace

3 Venice

meta data

duplicate both IDs and metadata

Distributing by **both** Columns and Rows (**Cells**)



56 has-name Marco Polo

ID	name
56	Marco Polo

12 has-name Niccolò Polo

ID	name
12	Niccolò Polo

35 has-name Maffeo Polo

ID	name
35	Maffeo Polo

server A

56 has-birth-date 1254

ID	birthDate
56	1254

12 has-birth-date 1230

ID	birthDate
12	1230

56 has-birth-place Venice

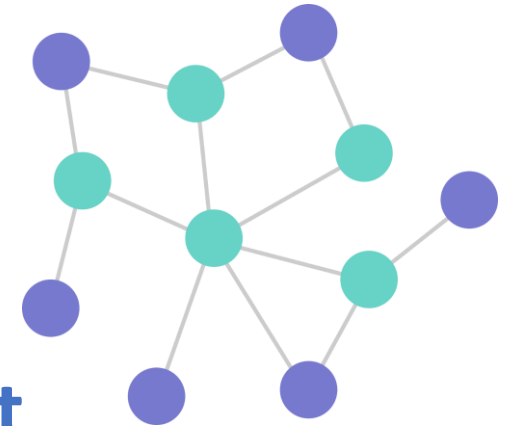
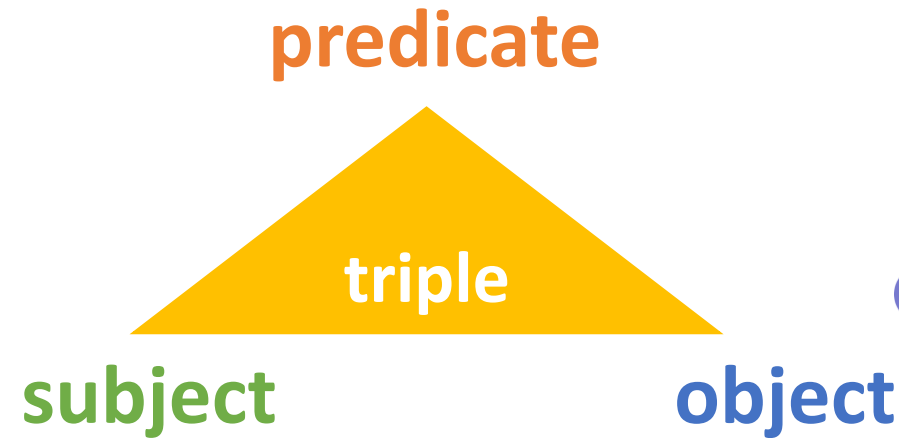
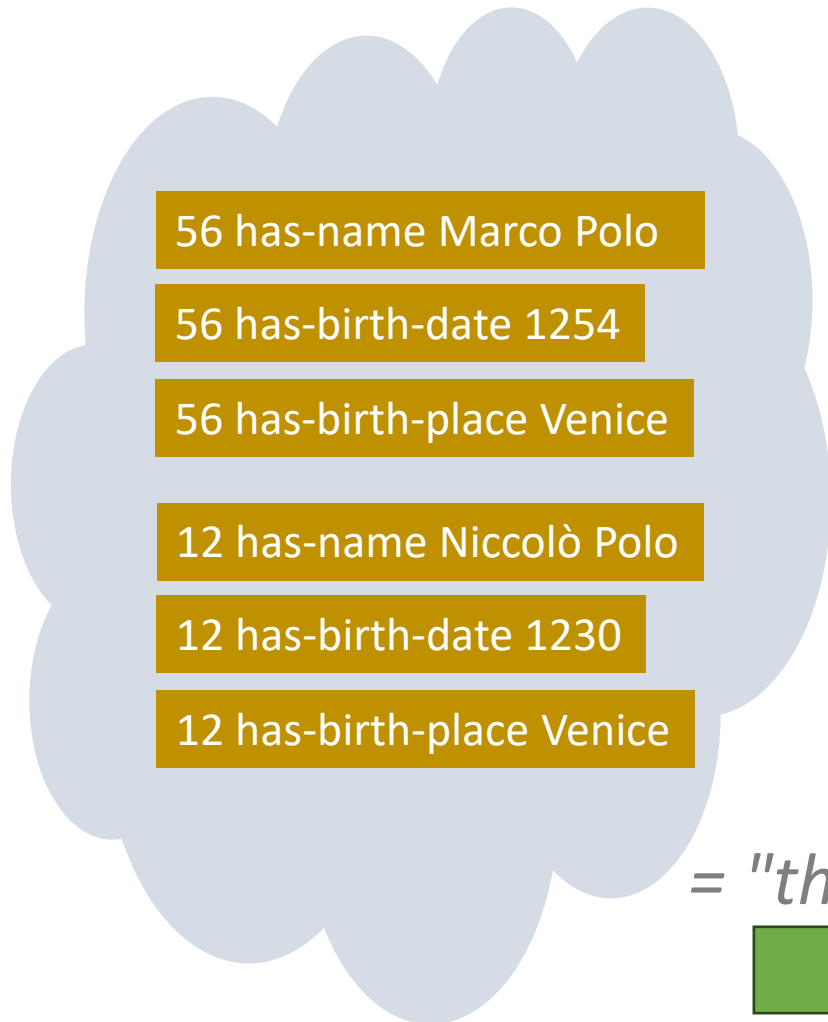
ID	birthPlace
56	Venice

12 has-birth-place Venice

ID	birthPlace
12	Venice

server B

Breaking Out of the Cage: Triple and RDF Data Model

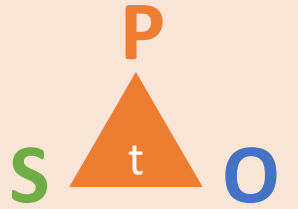


1. Subject (56) = row
 2. Predicate (has-name) = column
 3. Object = literal value (Marco Polo)
- = "the entity identified by 56 has name Marco Polo"

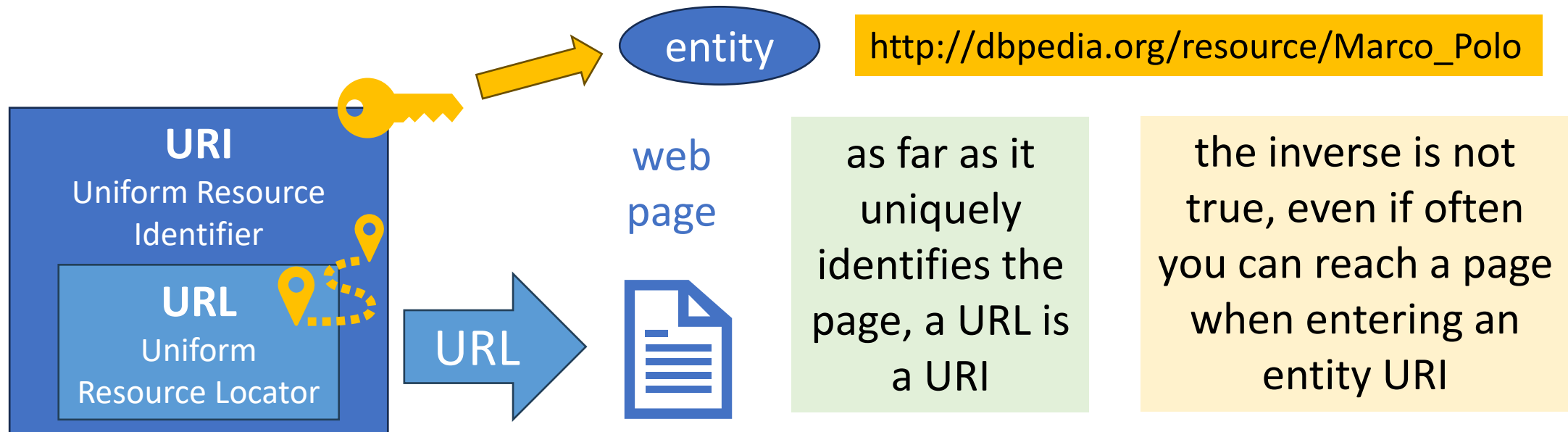
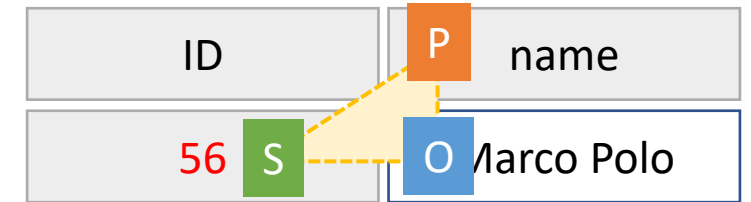
Subject	Predicate	Object
---------	-----------	--------

how to provide globally unique identifiers worldwide to identify each entity?

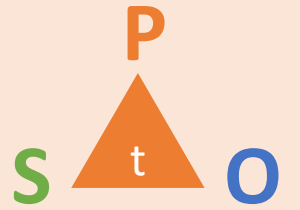
Modeling Data in RDF: URIs as Identifiers



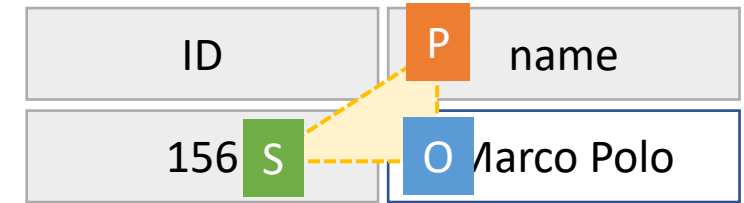
- in a global schema, each part of the triple (S,P,O) must be uniquely **identified**
- no central authority for an **open** web
- solution from **existing** web technologies



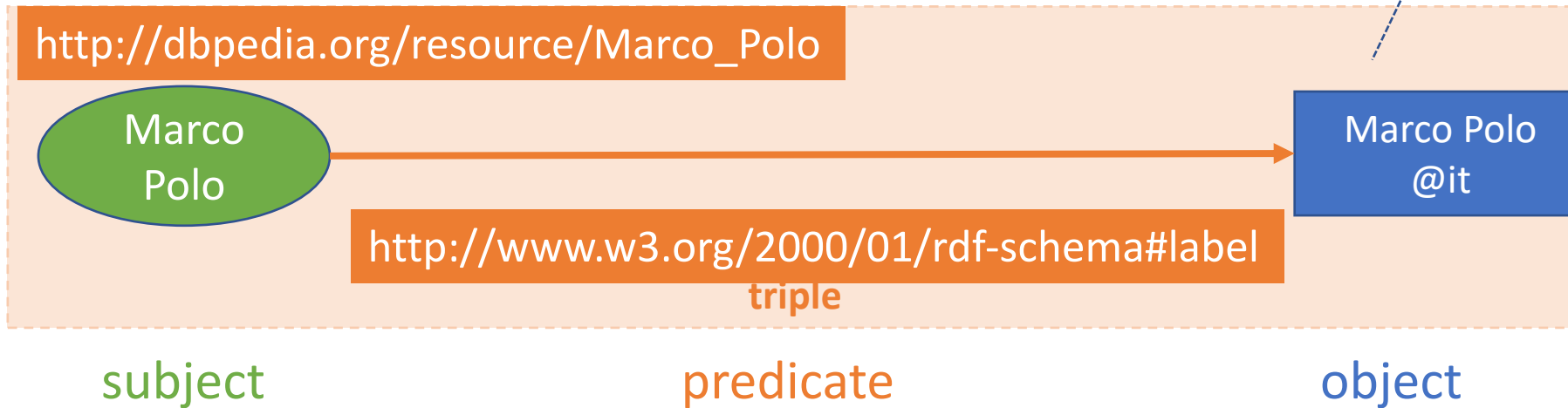
Modeling Data in RDF: Identifiers Usage



- each part of the triple (**S**, **P**, **O**) is identified by an URI except for **literal** objects (simple values like string, number, date...)
- literals often have a **type** specifier



LITERAL type = string
language = Italian



any data is expressed by triples: e.g. *Marco-Polo...*

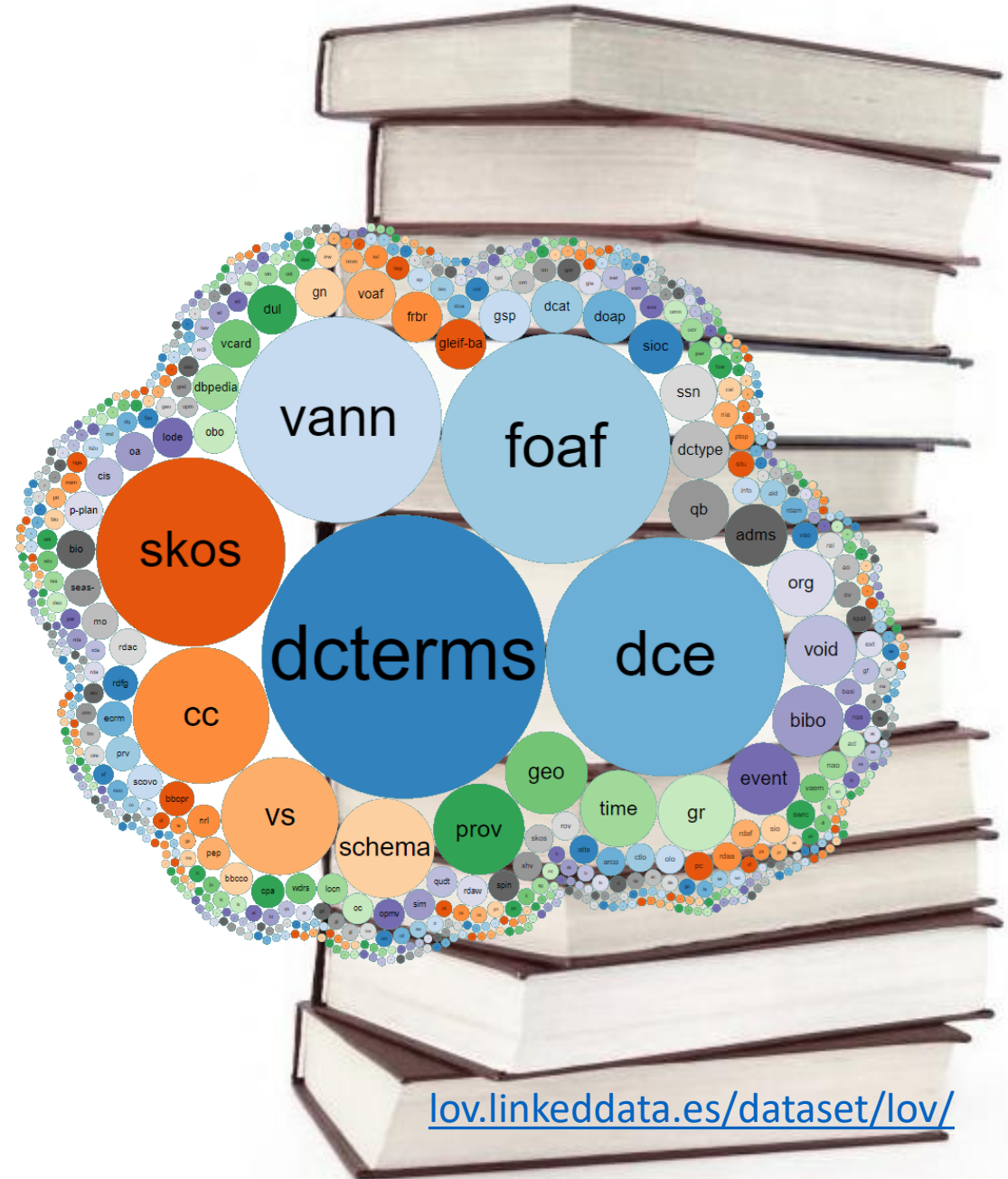
- *has-father* Niccolò-Polo;
- *has-birth-date* 1254;
- *has-birth-place* Venice ...

entity has-name "Marco Polo" (Italian)

this implies having **vocabularies** defining all the resources used as **S**, **P**, **O**, each with its own URI

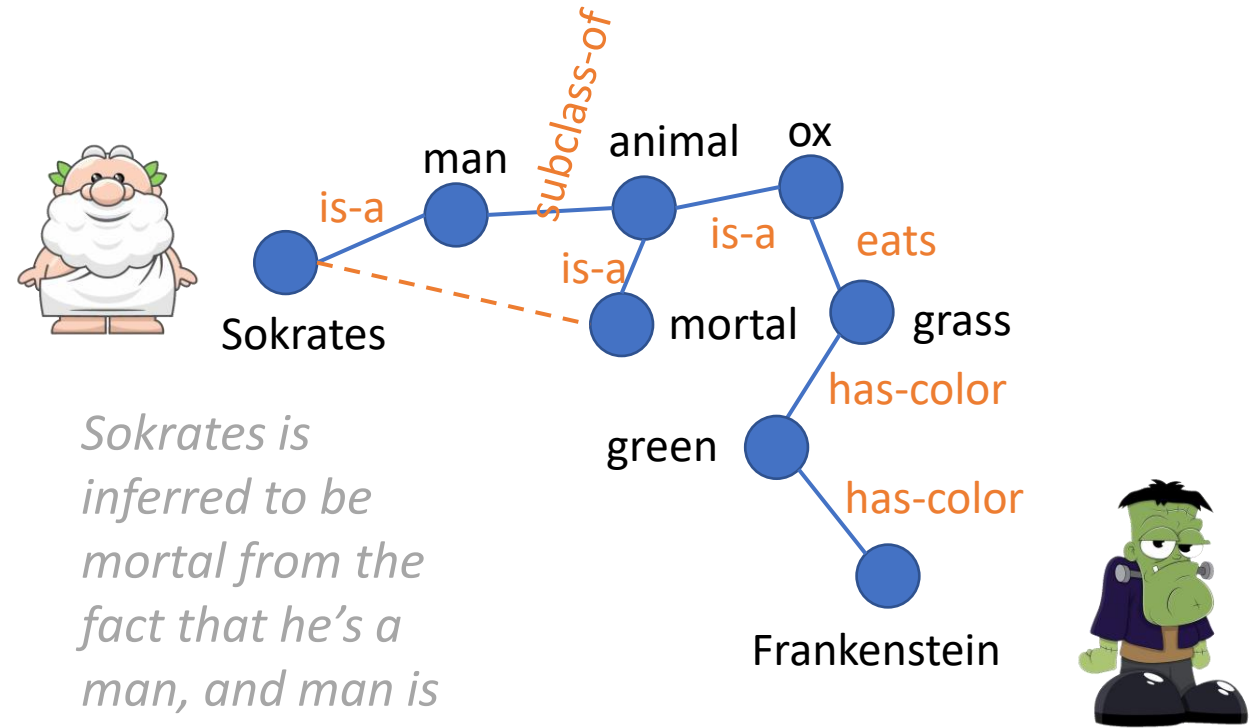
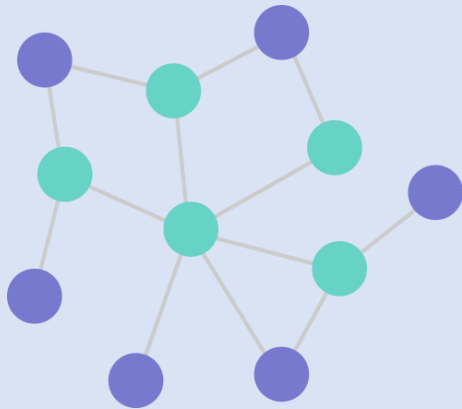
A World of Vocabularies

- several **vocabularies** are used to build *ontologies* (set of concepts and their relations):
 - **general-purpose** (e.g. **Dublin Core** for generic metadata: title, description, creator, date...)
 - **concept-specific**, e.g.:
 - **FOAF** (*friend of a friend*): persons, their activities and relations
 - **DBpedia** from Wikipedia infoboxes
 - **Schema.org** by major search engines (general-purpose metadata for web sites)
- anyone can create vocabularies, yet trying to **reuse** concepts from others



The Graph: a Global Network of Data

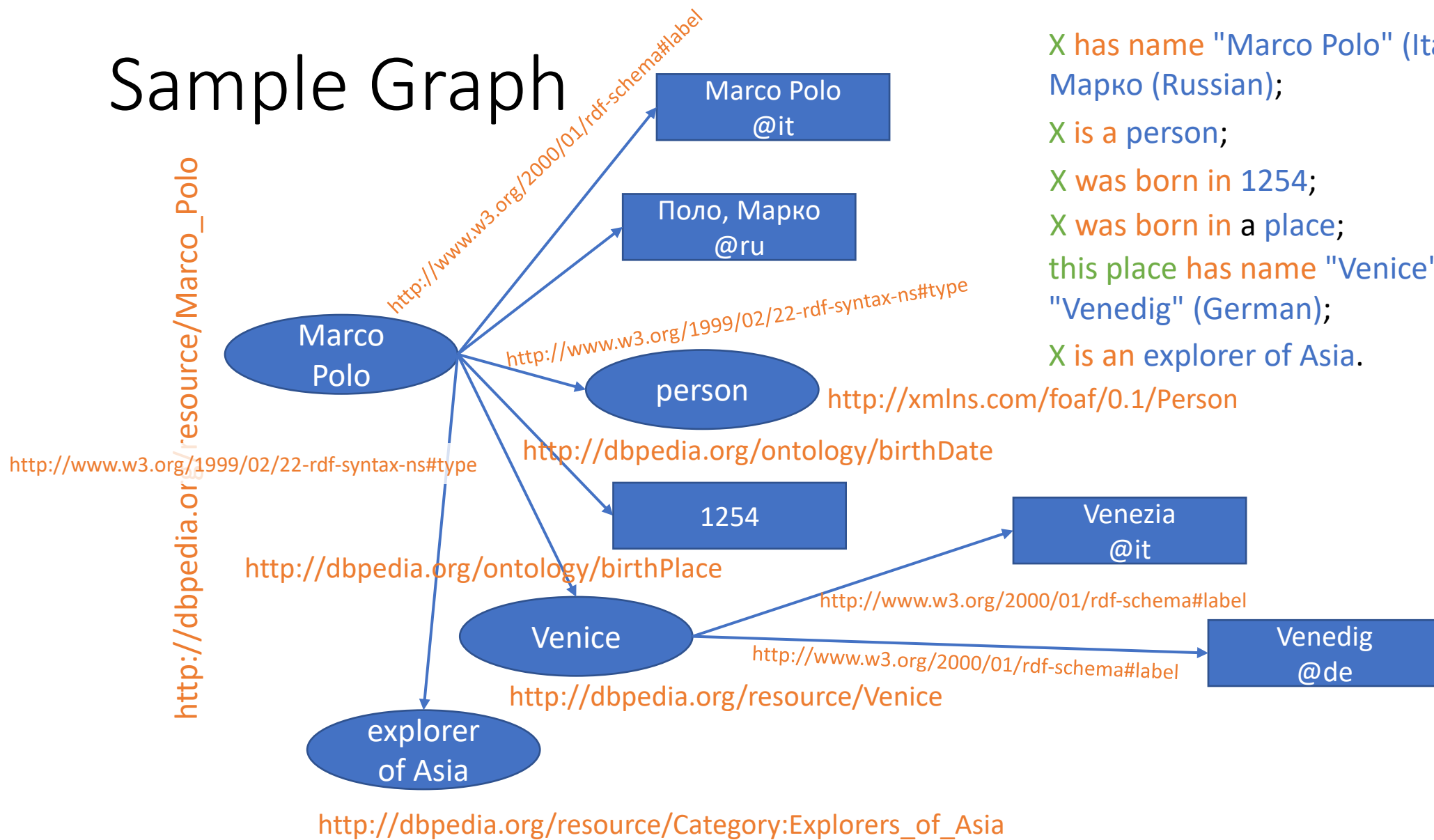
- triples can be connected each other without limits, building a giant **graph**
- by **walking** the graph we can surf knowledge domains and even let machines **infer** data



Sokrates is inferred to be mortal from the fact that he's a man, and man is a subclass of animals, and animals are told to be mortal.

we walked seamlessly from Sokrates to Frankenstein, via green things like grass, which are eaten by oxen, which are animals like men, Sokrates being a man...

Sample Graph

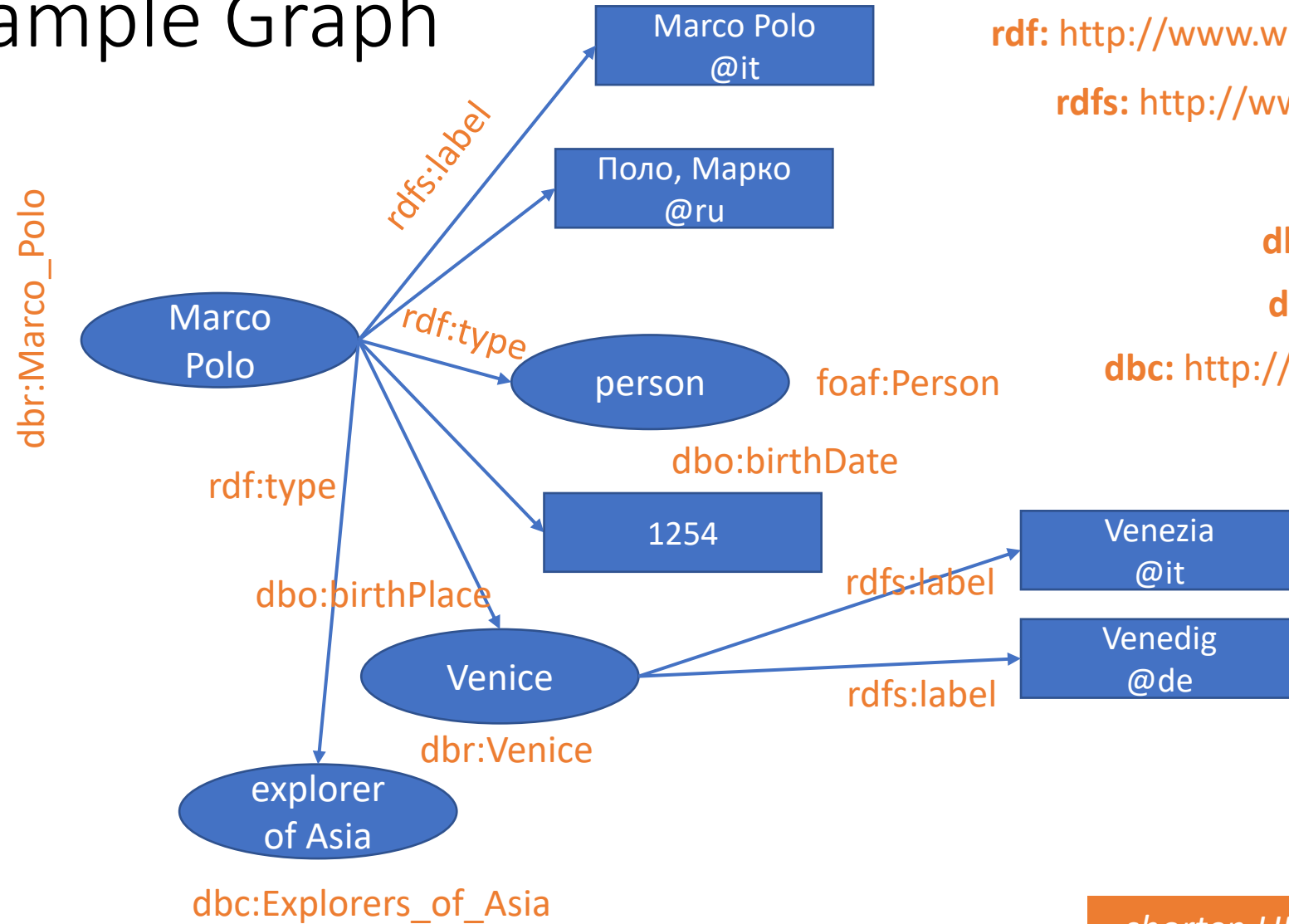


X has name "Marco Polo" (Italian), Поло, Марко (Russian);
X is a person;
X was born in 1254;
X was born in a place;
this place has name "Venice" (English), "Venedig" (German);
X is an explorer of Asia.

everything is a triple, and each part in it has its own URI (except for literals)

... URIs are verbose!

Sample Graph



rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

rdfs: <http://www.w3.org/2000/01/rdf-schema#>

foaf: <http://xmlns.com/foaf/0.1/>

dbo: <http://dbpedia.org/ontology/>

dbr: <http://dbpedia.org/resource/>

dbc: <http://dbpedia.org/resource/Category/>

shorten URIs by replacing their first portion with an arbitrarily chosen prefix (qnames)

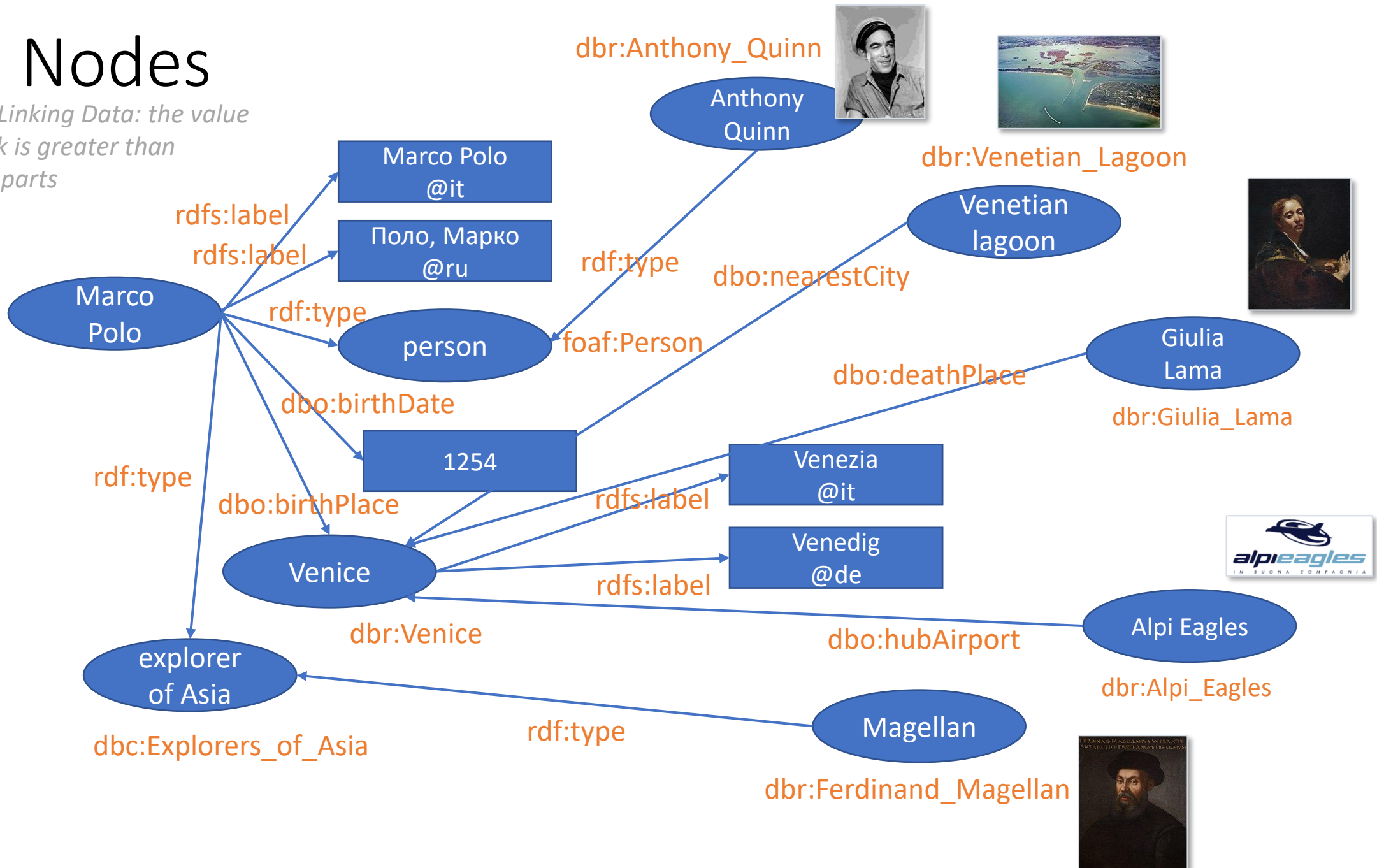
qualified names (qnames)

More Nodes

The power of Linking Data: the value of the network is greater than the sum of its parts

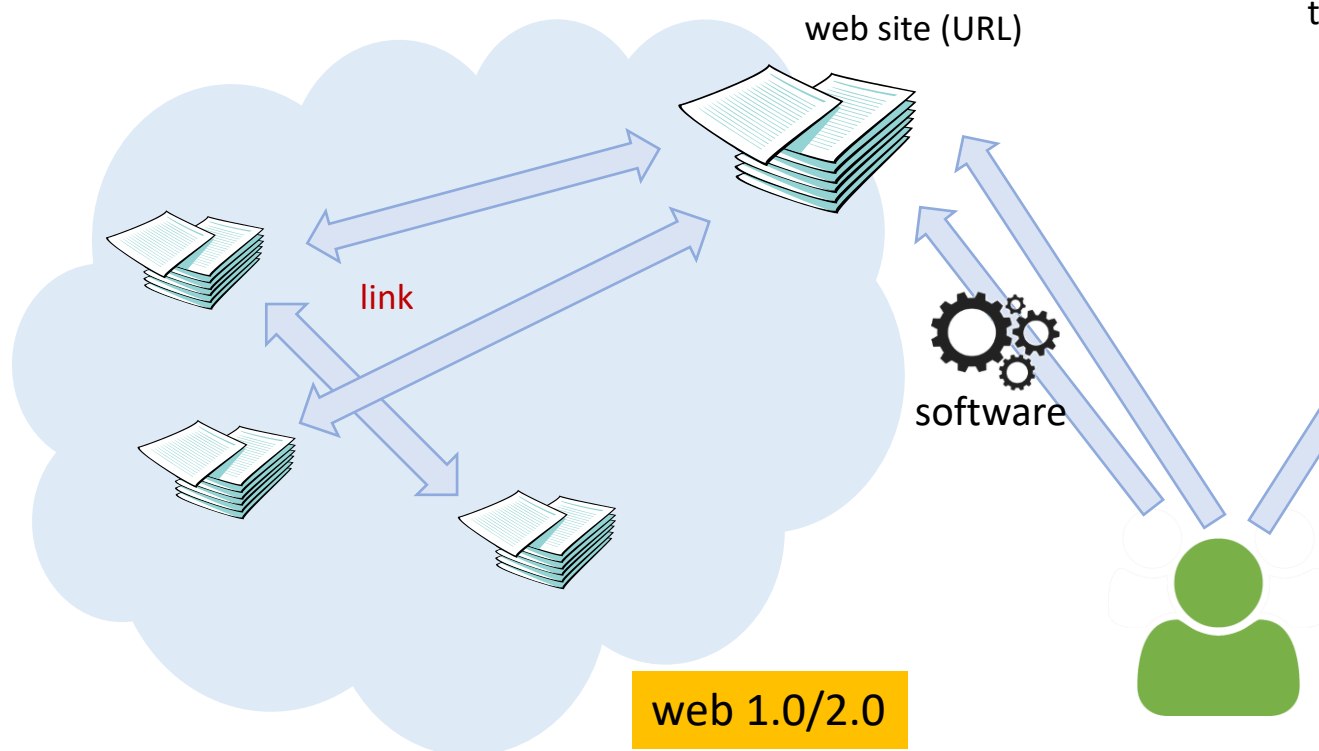


dbr:Marco_Polo



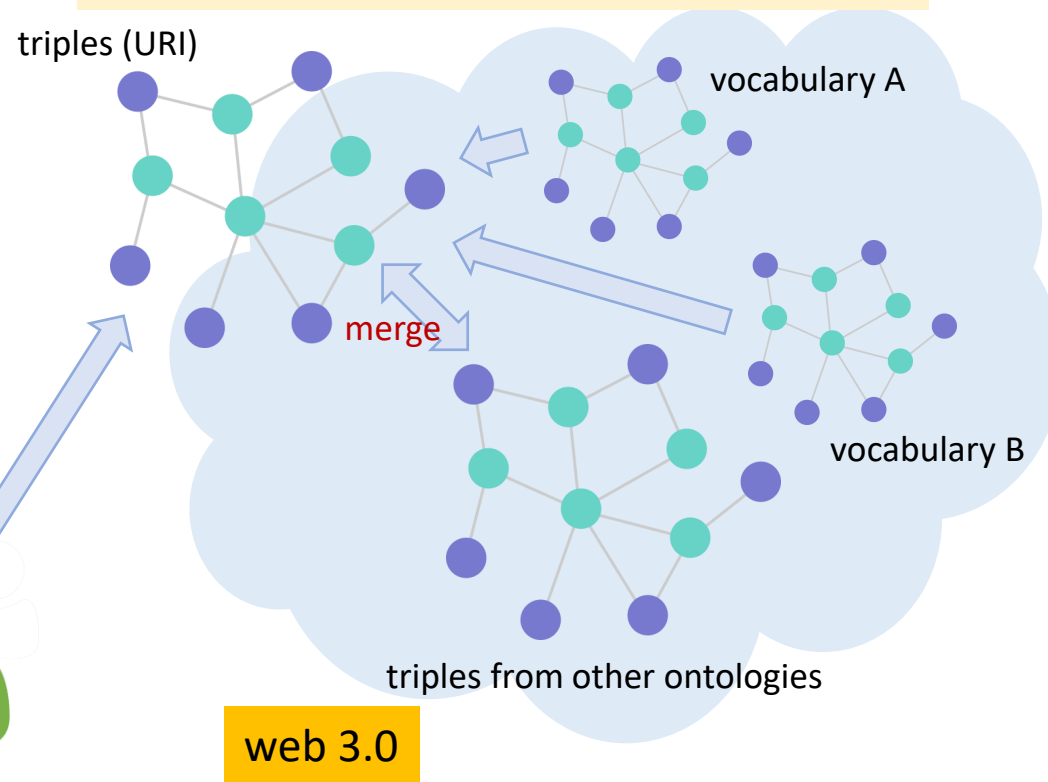
Publishing Data on the Web

publishing data **presentations** (however generated) as **hypertexts/GUIs** for **humans**



*anyone can publish pages in a site, which gets hyperlinked to other sites.
Pages are identified by URLs.*

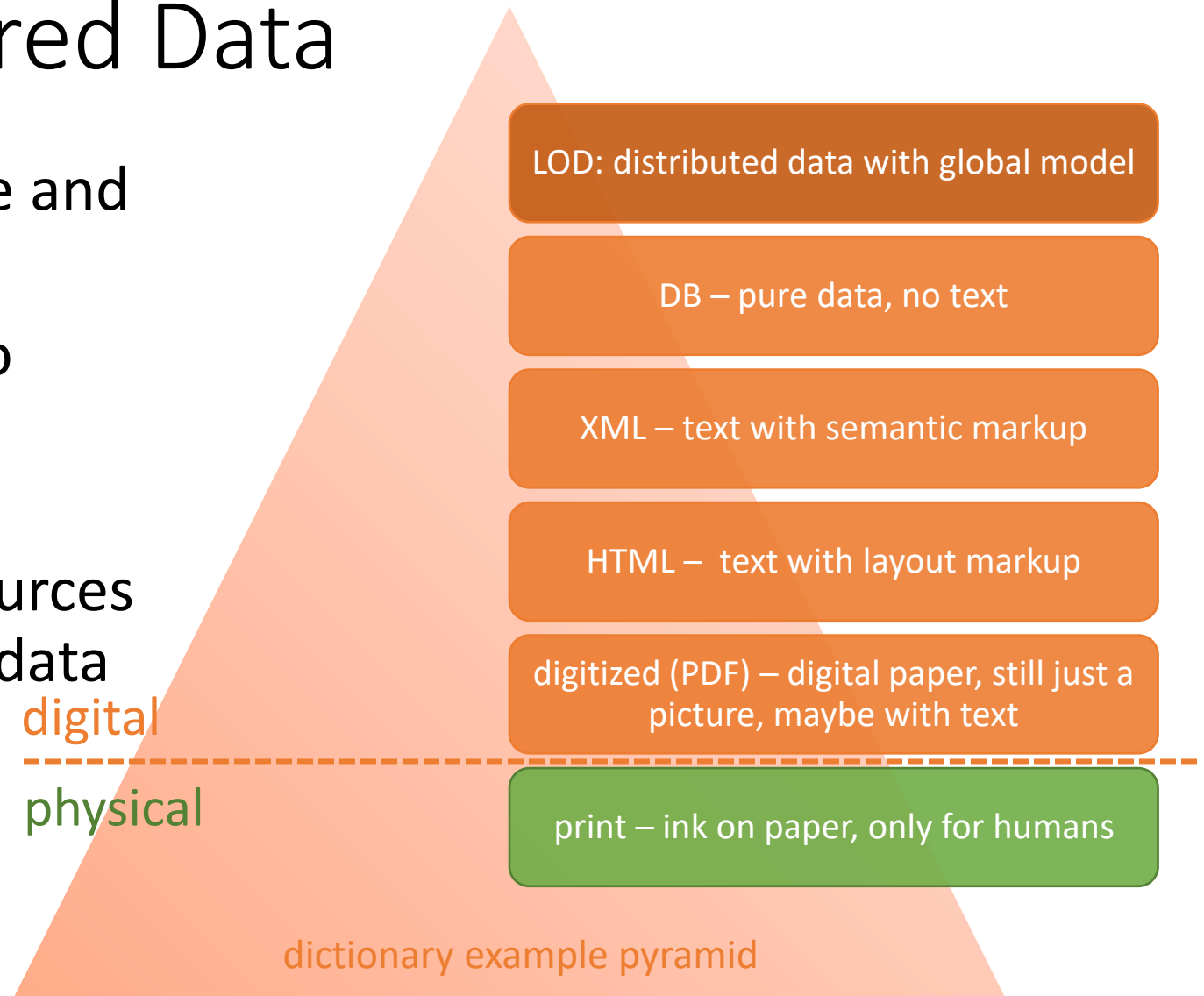
publishing data **representations** as **triples** for **machines** (RDF)



*anyone can publish data as triples, which get merged into the global data graph.
Concepts are identified by URIs.*

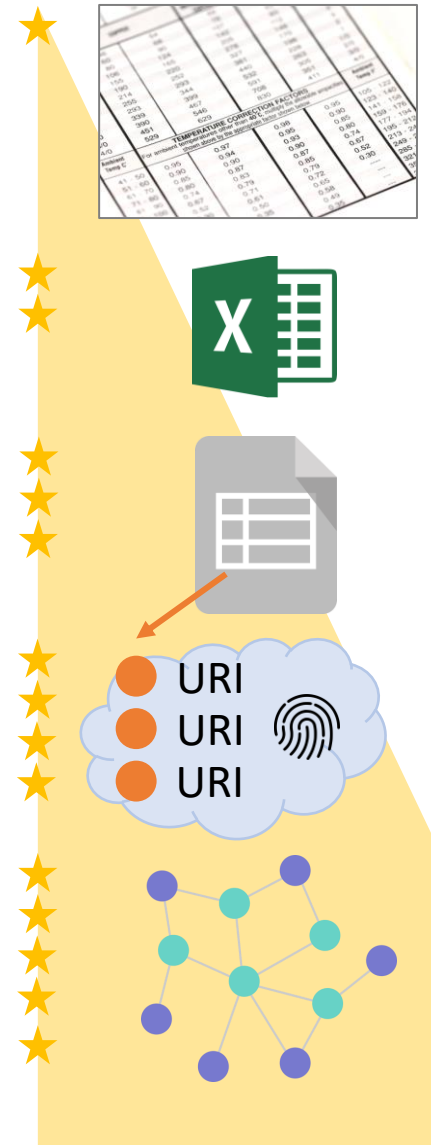
Distributing Structured Data

- comparable level of structure and granularity
- shared global model, open to indefinite integration and expansion
- often, XML or DB based resources can project a subset of their data into LOD



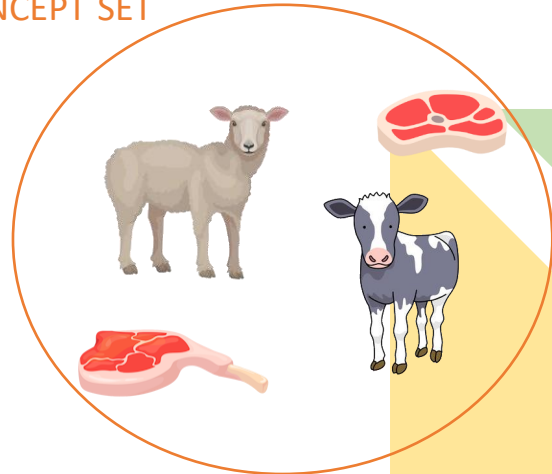
5-Stars Evaluation of Published Data

- 1 star: **available on the web** (e.g. image of a printed table)
 - *data locked in the document.*
- 2 stars: **machine-readable structured format** (e.g. XLS)
 - *data locked in the proprietary format.*
- 3 stars: **standard format** (e.g. CSV)
 - *data unlocked but on rather than in the web.*
- 4 stars: **global identifiers** (URI like URL)
 - *data in the web.*
- 5 stars: **linked data**
 - *data in the web and linked to other data.*

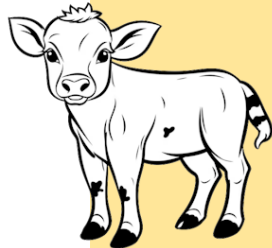


LEMON (Lexicon Model for Ontologies)

CONCEPT SET



LEXICAL CONCEPT



LEXICAL SENSE



LEXICAL SENSE



ONTOLOGY ENTITY



meanings

GRAPHIC FORM

vitello

PHONETIC FORM

/vit'ello/

LEXICAL ENTRY

“vitello”

morphological expressions

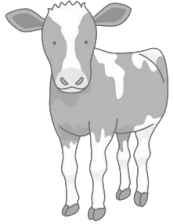


“calf”



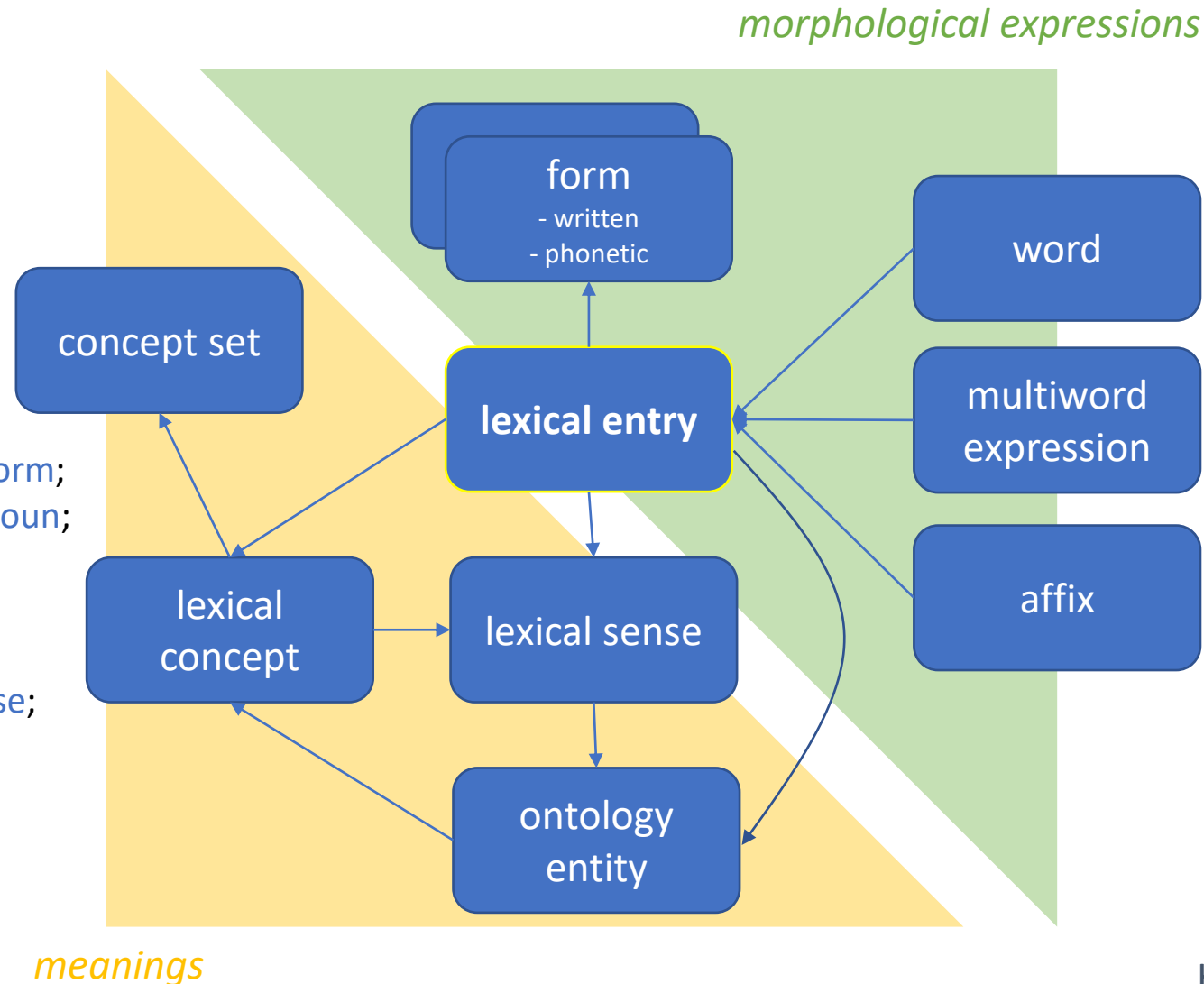
“veal”

The Lemon Core Lexical Ontology



A calf in Lemon Triples

```
x:calf a lemon:LexicalEntry;  
  lemon:canonicalForm x:calf_form;  
  lemon:partOfSpeech lexinfo:noun;  
  lemon:sense x:calf_sense.  
x:calf_form a lemon:Form;  
  lemon:writtenRep "calf"@en.  
x:calf_sense a lemon:LexicalSense;  
  lemon:reference  
<http://mywords.org#Calf>.
```



<https://lila-erc.eu/query>



Storing Data

A Glimpse at Turtle



Serializing Data

- RDF is an **abstract** data model; we need some serialization format to store its triples
- several **formats**, targeted to different typical usages, like:
 - **RDF/XML** (machine to machine)
 - **JSON-LD** (machine to machine)
 - **N-Triples** (verbose)
 - **Turtle** (and **TriG** for multiple graphs)



<https://www.w3.org/TR/turtle>

Serialization: Turtle - Tokens



- URIs in `<>`
- qnames: `prefix:name` where *prefix* is an initial portion of the URI
- prefixes are defined in a preamble:
 - `@prefix PREFIX: <URI>`.
 - `@base PREFIX: <URI>`. default prefix (e.g. `:name`)
- literals wrapped in `""`
- literal type (optional): `^^type` (usually XSDL datatype)
- abbreviation `a` = `rdf:type`

QNames

```
@PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>.
@PREFIX foaf: <http://xmlns.com/foaf/0.1/>.
@PREFIX dbr: <http://dbpedia.org/resource/>.
```

```
dbr:Marco_Polo rdf:type foaf:Person.
```

Literals

```
"123"^^xs:integer
```

```
"1254-6-28"^^xs:date
```

Abbreviations

```
dbr:Marco_Polo a foaf:Person.
```



Serialization: Turtle - Statements



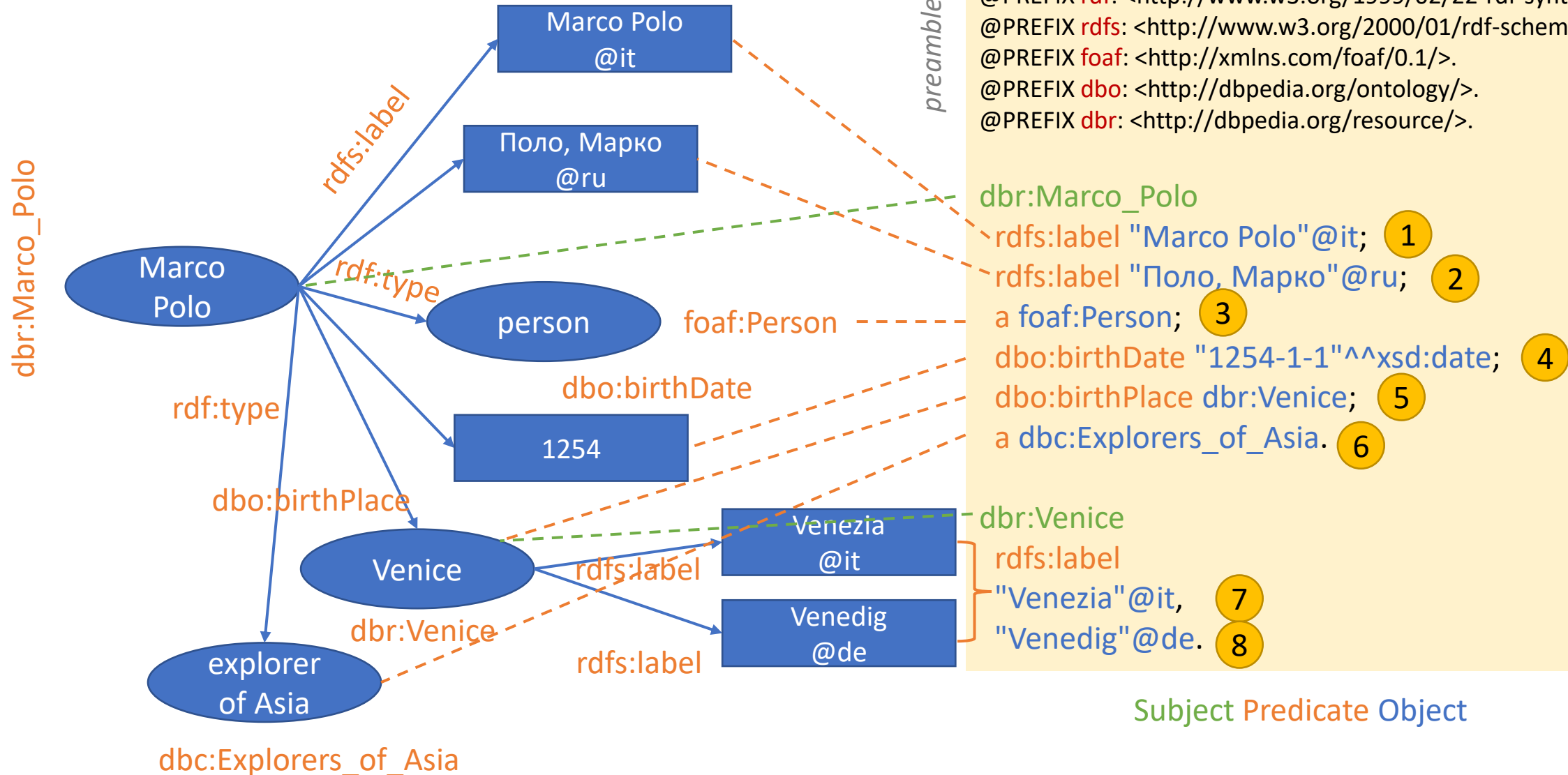
- statements **ended** by .
- multiple statements about the **same subject**: all but last ended by ;
- multiple statements about the **same subject and predicate**: all but last ended by ,

```
@PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>.
@PREFIX foaf: <http://xmlns.com/foaf/0.1/>.
@PREFIX dbr: <http://dbpedia.org/resource/>.
```

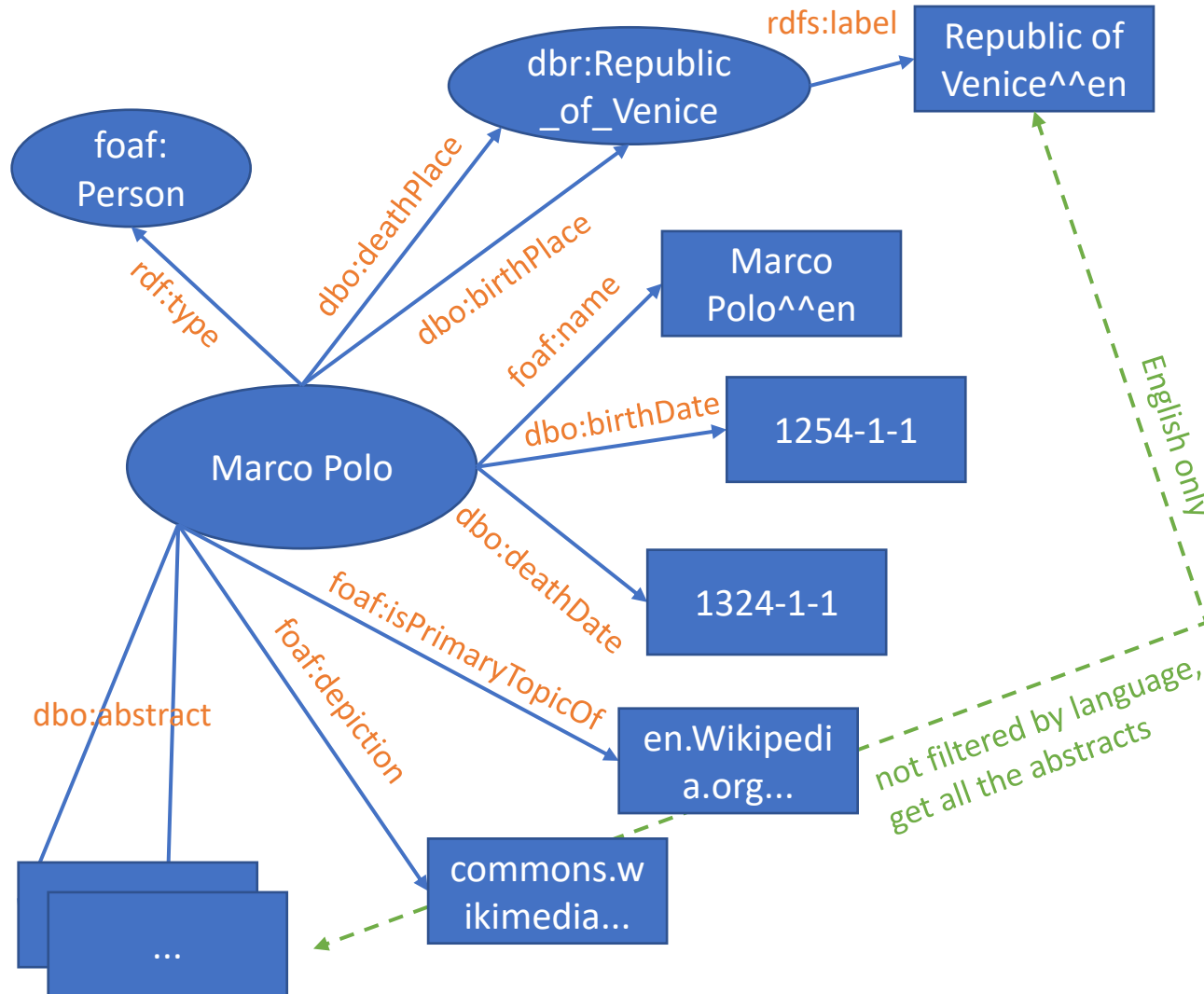
```
dbr:Marco_Polo
rdfs:label "Marco Polo"@it;
rdfs:label "Поло, Марко"@ru;
a foaf:Person.
```

3 triples, shared
subject

Turtle Sample



Querying LOD Data: SPARQL



```

PREFIX dbp: <http://dbpedia.org/property/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX foaf: <http://xmlns.com/foaf/0.1/>
PREFIX dbr: <http://dbpedia.org/resource/>
SELECT DISTINCT dbr:Marco_Polo as ?person ?name
    ?birth_date ?birth_place ?birth_place_label
    ?death_date ?death_place ?death_place_label
    ?topic ?depiction ?abstract
WHERE {
    dbr:Marco_Polo a foaf:Person;      must be a person
    foaf:name ?name.                  having a name
    OPTIONAL {
        dbr:Marco_Polo dbo:birthDate ?birth_date;
        dbo:deathDate ?death_date;    if present, we also want:
        foaf:isPrimaryTopicOf ?topic; birth/death date, wiki
        foaf:depiction ?depiction;    page, default picture,
        dbo:abstract ?abstract.       abstract...

        dbr:Marco_Polo dbo:birthPlace ?birth_place. ... birth place
        ?birth_place rdfs:label ?birth_place_label. and its label...

        dbr:Marco_Polo dbo:deathPlace ?death_place. ... death place
        ?death_place rdfs:label ?death_place_label. and its label...
    }
    FILTER(lang(?birth_place_label)="en") only English labels
    FILTER(lang(?death_place_label)="en") for places
}
    
```



Practicing LOD



LOD: Benefits and Limitations

Benefits

- **universal model** designed to link data worldwide
- **lots of ontologies** usable as building blocks
- **pure content**, language-independent, built for **machines** and allowing **reasoning**, as a **deterministic complement** to AI probabilistic approach
- based on **existing web techs**

Limitations

- **scalability**: the performance penalty becomes a show-stopper beyond a certain size
- **open world**, constantly changing and potentially breaking software
- **content creation**: often data is not created directly as RDF triples, too fragmented for our think habits

Data Particles and Higher-Level View

“Petrarch was born in Arezzo
on July 20, 1304 from ser
Petracco and Eletta Canigiani”

x:events/12/birth	rdf:type	crm:e67_birth
x:events/12/birth	crm:p2_has_type	x:event-types/birth
x:events/12/birth	crm:p98_brought into life	x:people/petrarch
x:events/12/birth	crm:p3_has_note	born 1304/7/20 from ser Petracco and Eletta Canigiani

x:places/arezzo	rdf:type	crm:e53_place
x:events/12/birth	crm:p7_took_place_at	x:places/arezzo
x:events/12/birth	crm:p4_has_timespan	x:timespans/ts#11
x:timespans/ts#11	crm:p82_at_some_time_within	1304
x:timespans/ts#11	crm:p87_is_identified_by	20 Jul 1304 AD
x:events/12/birth	crm:p96_by_mother	x:people/canigiani
x:events/12/birth	crm:p97_from_father	x:people/petracco

Integrating LOD

